

Saif Shikalgar

Embedded Systems & Firmware Engineer

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Education

G. H. Raisoni College of Engineering and Management

Pune, India

B.Tech in Computer Science & Engineering (Cybersecurity Specialization); **CGPA: 8.10/10.0**

Sep 2023 – May 2027 (Expected)

Coursework: Microcontrollers & Embedded Systems, Computer Networks, Operating Systems, Real-Time Systems, Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS.

Technical Skills

Microcontrollers & Architectures: STM32 (ARM Cortex-M4, STM32F446RE), ESP32 (Xtensa Dual-Core), ESP8266, Arduino AVR, NVIDIA Jetson Nano, Register-Level Direct Hardware Programming.

Firmware, RTOS & Protocols: Bare-Metal C, C++ (C11/C++20), FreeRTOS (Task Scheduling, Queues, Semaphores), CAN Bus 2.0B / TWAI, OBD-II, SPI, I2C, UART, PWM, BLE GATT, DMA Controllers, ISR Optimization.

Hardware Tools & Debugging: STM32CubeIDE, ESP-IDF, Keil uVision, Logic Analyzers, Digital Oscilloscopes, JTAG/SWD Debugging, Multimeters, Wireshark, MCP2515 CAN Transceivers.

Languages & Tooling: C, C++, Python, Bash Scripting, Linux (Embedded Ubuntu/Debian), Git, Docker.

Work Experience & Engineering Leadership

Team Abhyuday Racing

Pune, India

Technical Lead & Department Head – Drive-by-Wire (Automotive R&D)

Jan 2025 – Present

- Designed and deployed a distributed **3-ECU network (NVIDIA Jetson Nano + STM32F446RE + ESP32)** over CAN 2.0B bus with hardware heartbeats, priority arbitration, and 99.9% packet reliability under high EMI.
- Programmed real-time deterministic **Autonomous Emergency Braking (AEB)** control logic, securing **1st Place Nationally at aBAJA 2026** by halting safely at **6.2m** at 30 km/h with **40 bar hydraulic line pressure**.
- Spearheaded an emergency **8-hour overnight hardware migration** from damaged STM32 to ESP32 FreeRTOS architecture at the racetrack, winning **1st Place in Adaptive Cruise Control (ACC)** at aBAJA 2025.
- Engineered a modular Brake-by-Wire linear actuator system mounted parallel to the tandem master cylinder, cutting costs by **50%** and winning the **1st Place Manufacturing Excellence Award**.
- Optimized low-level driver interrupt handlers and sensor-actuator feedback loops on NVIDIA Jetson, eliminating CAN frame drop bottlenecks.

AuraByte Studios (Independent Venture)

Pune, India

Lead Embedded Systems Developer & Founder

Dec 2025 – Present

- Engineered low-level protocol adapters and mathematical modeling engines handling complex fiber network calculations for over **2.09K+ active users**.

Flagship Embedded Systems Projects

Dual-Gate STM32 Hardware CAN Firewall & Anomaly IDS/IPS | STM32F446RE, C11, CMSIS/HAL, FreeRTOS, CAN 2.0B

- Engineered an inline dual-transceiver CAN bus firewall intercepting OBD-II injection, fuzzing, and RPM spoofing attacks with a deterministic **5.92 μ s processing latency** (1,065 clock cycles at 180 MHz).
- Implemented fixed-point Mahalanobis distance anomaly detection and jitter statistical analysis, rejecting **88.5% of noise** and achieving a **99%+ threat mitigation rate** with zero bus contention.
- Leveraged direct register manipulation and DMA ring buffers on STM32F446RE to minimize ISR context-switching overhead.

24S LiFePO4 Battery Management System (BMS) Telematics Platform | ESP32, FreeRTOS, BLE GATT, Python, C++

- Reverse-engineered industrial JK BMS BLE GATT protocols to extract real-time cell voltage deltas (<15mV), current draws, and temperatures across a 76.8V 24S LiFePO4 battery pack with automated thermal runaway cutoff.
- Implemented multi-threaded FreeRTOS task architecture separating high-speed ADC sensor sampling from wireless telemetry broadcasting.

Suraksha360: Hardware Security Scanner | Arduino Pro Micro, ESP8266, C++, Hardware USB HID

- Engineered a plug-and-play USB hardware penetration testing tool that executes automated zero-install system configuration analysis upon connection to host machines.

Honors & Awards

1st Place, Autonomous Emergency Braking (AEB): National aBAJA 2026 Competition; achieved a 6.2m halt on a 6.0m autonomous target threshold at 30 km/h.

1st Place, Manufacturing Excellence Award: National aBAJA 2026 Competition; recognized for modular Drive-by-Wire and Brake-by-Wire actuator design.

1st Place, Adaptive Cruise Control (ACC): aBAJA 2025; achieved following an 8-hour high-stakes crisis hardware migration from STM32 to ESP32.

1st Runner Up, MATLAB Advanced Simulation: National aBAJA 2026 Competition; developed a GPS-based Point-A to Point-B autonomous traversal algorithm.